

Mykola Zhuk

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📁 Experience

Software Developer - Contract

Jun 2026 – Present | Remote

B2B SaaS Startup

- Built a **Python pipeline** extracting 20 corporate-event types from 2,936 articles into 1,332 records, with **99.7% passing client-defined schema validation**.
- **Deployed and tested on Azure** using Container Apps Jobs, Service Bus, and Blob Storage, with PostgreSQL-backed idempotency, automated tests, and audit logs.
- Reduced **LLM calls by 17.6% and token usage by 17.9%** through caching and selective inference.
- Estimated operating costs of **US\$400/month for 30,000+ articles**, versus the founder's US\$6,000/month estimate for existing infrastructure and external services.

Backend Engineer - Volunteer

Oct 2025 – Mar 2026 | Remote

Tacmedbat

- Built NestJS/MongoDB APIs and a custom admin interface enabling the customer to edit Ukrainian/English landing-page content. Designed a public endpoint combining **8 collections to support 10+ frontend blocks**.
- Delivered **15+ backend features and fixes** across Cloudinary image handling, **role-based access control**, request validation, payments, and SMTP delivery.

Software Analyst

Jun 2024 – Aug 2024 | Toronto

Accenture

- Developed Python/OpenAI API prototypes for retail Q&A and semantic search, supported by data pipelines ingesting and enriching **50,000+ product records**.
- Delivered features with a cross-functional team through Agile sprints, code reviews, and stakeholder demos.

🔧 Projects

Full-Stack Engineer

Oct 2024 – Dec 2024 | Toronto

Scriptorium-CSC309 Project

- Built **20+ REST API endpoints** with **JWT-authenticated, role-based access** and React/Next.js flows for code forking, search, and execution.
- Built Docker language environments with **execution-time and memory limits for 10+ languages**. Deployed the application on AWS EC2.
- Added **GitHub and Google OAuth 2.0 sign-in**. Tested APIs with Postman and checked responsive layouts.
- **Live Demo (AWS)**

Backend Engineer

Oct 2024 – Dec 2024 | Toronto

ProjectOllon (Industry-Partnered MVP) - CSC301 Project

- Designed a Prisma schema for **100+ files and 2,000+ code components**, reducing full-scan time by **80%** in project evaluation.
- Built **static-analysis checks for style and duplication** and integrated OpenAI/Cohere models to flag code complexity; helped deliver the code-review MVP to industry partner Ollon.

Machine Learning Engineer

Mar 2025 – Apr 2025 | Toronto

ML Classifier- CSC311 Project (scikit-learn, Pandas, PyTorch)

- Compared Random Forest and PyTorch models on **1,644 survey rows**, with preprocessing and respondent-grouped splits.
- Achieved **89.96% accuracy on 249 held-out rows**. Packaged the model and preprocessing for CLI prediction.

🧠 Skills

Languages: Python, Java, JavaScript/TypeScript, SQL, Bash

Frameworks & Data: React, Next.js, Node.js, NestJS, PostgreSQL, MongoDB, Prisma, PyTorch, scikit-learn

Cloud & testing: Azure, AWS EC2, Docker, GitHub Actions, pytest, Jest, Grafana Loki, Postman, Git, Linux

🎓 Education

Honours B.Sc, Mathematics

Jun 2025

University of Toronto

Relevant Courses: Software Design, Machine Learning, Data Structures and Algorithms, Web Programming, System Programming, Computer Organization